

# Glossary

**absolute value function** a function whose rule contains an absolute value expression

**accuracy** how close a measure is to its true value

**acute angles** angles that measure less than  $90^\circ$

**addition counting principle** if there are  $m$  ways of doing one thing and  $n$  ways of doing another thing, then there are  $m + n$  ways of doing one thing or the other

**adjacent angles** angles that share a common side, have the same vertex, and do not share any interior common points

**adjacent side of acute angle in a right triangle** the side next to the given angle in a right triangle but is not the hypotenuse

**alternate exterior angles** angles outside the two lines that are not the transversal and are on diagonal opposite sides of the transversal

**altitude of a pyramid** a perpendicular line segment that joins the vertex to the plane of the base; An altitude may lie inside, on, or outside a pyramid.

**altitude of a triangle** a perpendicular segment from a vertex of the triangle to a line containing the base opposite the vertex

**amplitude of a sinusoidal function** one-half the distance between the maximum value and the minimum value; the height of a sinusoidal function above the midline

**analytic geometry** the study of geometry using the tools of algebra; also called coordinate geometry

**angle** a figure formed by two rays, called sides, that share the same endpoint

**angle bisector** a line or line segment or ray that divides an angle into two congruent angles

**angle of depression of an object** the downward angle that the object makes with the horizontal

**angle of elevation of an object** the upward angle that the object makes with the horizontal

**annulus** the region between two concentric circles

**arc** the set of points on a circle between two points on the circle

**arc length** a part of the circumference of a circle; It can be found by writing the part as a fraction over  $360^\circ$  and multiplying by the circumference.

**arccosine (arccos) function**  $y = \arccos x$  if and only if  $\cos y = x$  where  $-1 \leq x \leq 1$  and  $0 \leq y \leq \pi$

**arcsine (arcsin) function**  $y = \arcsin x$  if and only if  $\sin y = x$  where  $-1 \leq x \leq 1$  and  $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

**area** the number of square units contained in the interior of a figure

**argument** a set of statements, called premises, which are needed to reach a conclusion; Both the premise and the conclusion are considered to be part of the argument.

**arithmetic sequence** a sequence in which the difference between consecutive terms is a constant

**association** the relationship between two variables

**asymptote** a line that a graph of a given function approaches without touching

**average rate of change in a quantity** the ratio of the change in a quantity to the change in time

**axis of a cylinder** the line connecting the centers of the bases

**axis of a double-napped cone** the line about which a generating line revolves to produce a double-napped cone

**axis of symmetry** a line drawn through a figure so that one side is a reflection of the image on the opposite side; also called a line of symmetry

**base** a number raised to an exponent; For example, in  $5^2$ , 5 is the base.

**base edges of pyramid** edges that form the base of a pyramid; A base edge is formed when a lateral face and the base meet.

**bell-shaped distribution** a type of symmetric distribution of data sets that is nearly bell-shaped, where the mean and median are nearly equal

**bi-** prefix that means “two”

**biased sample** a sample that is not representative of its population

**binomial** a polynomial with two terms

**bivariate data** data that show the relationship between paired variables

**boundary line** a line that divides the coordinate plane into two half-planes

**bounded closed interval** an interval that includes both endpoints

**bounded half-open interval** an interval that includes one and only one endpoint

**bounded interval** the set of all real numbers between two numbers, called endpoints; The endpoints may or may not be included.

**bounded open interval** an interval that does not include either endpoint

**box-and-whisker plot** a diagram that shows the distribution or spread of data with the minimum, the maximum, and the three quartiles of the data; The box extends from  $Q_1$  to  $Q_3$ . The median is on the vertical line in the box. The whiskers extend from the first quartile to the minimum and from the third quartile to the maximum.

**ceiling function** another name for the least integer function

**center of a circle** the point inside a circle that is an equal distance from every point on the circle

**center of a polygon** the point inside a polygon that is equidistant from each vertex

**center of a sphere** the point inside a sphere from which all points on the surface of the sphere are an equal distance

**center of dilation** the point where the lines connecting each point on the pre-image with its corresponding point on the image intersect

**central angle of a circle** an angle that has its vertex at the center of a circle and its sides as radii of the circle

**central angle of a polygon** an angle formed by line segments drawn from the center to two consecutive vertices

**central limit theorem** when several random samples of size  $n$  are taken from a population and  $n$  is sufficiently large, then the means of the samples will have a distribution that is approximately normal

**centroid** the point where all three medians of a triangle intersect

**chord** a segment that connects any two points on a circle

**circle** the set of all points in a plane that are a fixed distance  $r$  (the radius) from a given point (the center)

**circle (as a conic section)** a conic section formed when a plane intersects only one nappe, perpendicular to the axis

**circle graph** a graph that uses sectors of a circle to display and compare data that can be broken down into separate categories

**circumcenter** the point where the perpendicular bisectors on each side of a triangle intersect

**circumscribed circle** for a given triangle, the circle that contains each vertex of the triangle; The circumcenter of the triangle is the center of the circumscribed circle.

**classes** equal-sized groups used to separate and categorize data; also called intervals

**closed half-plane** a half-plane in a nonstrict inequality where the boundary line is a solid line

**closed set** a set such that, under an operation, the result of the operation on any two elements of the set is also an element of the set

**coefficient** the nonvariable factor of a term

**coefficient of determination** the square of the correlation coefficient; written  $r^2$

**coincident system** a system with infinitely many solutions

**collinear** points that lie on the same line

**common difference** the difference between a term and the previous term; the constant difference  $d$  in an arithmetic sequence such that  $d = a_n - a_{n-1}$

**common factor** a factor shared by two or more numbers

**common logarithm** a logarithm with base 10; Common logarithms, such as  $\log_{10} x$ , are usually written without the base, as  $\log x$ .

**common logarithmic function with base 10**  
 $f(x) = \log x$  when  $x$  is a positive real number; The domain is  $x > 0$  and the vertical asymptote is  $x = 0$ .

**common ratio of a geometric sequence** the ratio of a term to the previous term; the constant  $r$  in a geometric sequence such that  $r = \frac{a_n}{a_{n-1}}$

**complementary angles** angles whose measures sum to  $90^\circ$

**completing the square** the process of transforming an expression of the form  $x^2 + bx$  into a perfect square trinomial by adding the term  $\left(\frac{b}{2}\right)^2$  to it

**complex conjugates** two complex numbers of the form  $a + bi$  and  $a - bi$

**complex numbers** numbers of the form  $a + bi$  where  $a$  is the real part and  $b$  is the imaginary part and  $a$  and  $b$  are real numbers; denoted  $\mathbb{C}$

**complex plane** the plane on which every complex number can be graphed; The horizontal axis is the real axis and the vertical axis is the imaginary axis.

**compound event** an event that consists of two or more simple events

**compound inequality** a pair of inequalities joined by the word *and* or the word *or*; a type of compound statement

**computer algebra system (CAS)** a computer program that performs algebraic manipulations such as simplifying algebraic expressions and solving equations

**conclusion of a conditional statement** the part of a conditional statement that includes the words following *then*

**conclusion of an argument** the end part of an argument that follows the premises

**conditional probability** the probability of an event given that another event has already happened; The conditional probability that event  $B$  occurs after event  $A$  has already occurred can be represented as  $P(B | A)$ , which is read as “the probability of  $B$  given  $A$ .”

**conditional statement** a statement that has two parts; The first part begins with the word *if* and the second part begins with the word *then*.

**cone** a solid with a circular base, a vertex, and a curved surface

**confidence interval** an interval estimate that is likely to contain a population parameter being estimated

**confidence level** the percent chance that a population parameter falls inside the confidence interval

**congruent line segments** line segments that have equal length

**congruent polygons** polygons that are the same size and shape

**conic section** a two-dimensional graph that can be formed by the intersection of a plane with a double-napped cone

**conjecture** a statement that is thought to be true but is yet to be proven

**conjugate binomials** two binomials with the same terms but opposite signs; for example,  $(a + b)$  and  $(a - b)$

**conjugate pair** a pair of expressions of the form  $a + bi$  and  $a - bi$

**conjunction** a compound statement that uses the word *and*

**consistent dependent system of equations**  
a system with infinitely many solutions

**consistent independent system of equations**  
a system with exactly one solution

**consistent system of equations** a system with exactly one solution or infinitely many solutions

**constant** a term that has no variables

**constant function** a function is constant within an interval if the value of  $f(x)$  remains the same

**constraint** a necessary condition in a problem, often written in the form of an inequality

**continuous function** a function with a connected graph

**continuous random variable** a variable whose outcomes are both random and continuous

**contraction** a dilation for which the absolute value of the scale factor is between 0 and 1

**contradiction (equation)** an equation that is true for no values of the variable; A contradiction has no solutions, represented by the null set:  $\{ \}$  or  $\emptyset$ .

**contradiction (statement)** a statement that is logically at odds with a previous statement that was assumed to be true

**converse** a conditional statement that switches the hypothesis and the conclusion of the original conditional statement

**conversion factor** a fraction that is used to convert measures and rates and that has a numerator and denominator of the same quantity written in different units

**coordinate** the number on a number line that gives the location of a point

**coordinate plane** a plane in which the coordinates of any point are the point's distances from two intersecting perpendicular lines called axes (the  $x$ -axis and  $y$ -axis)

**coplanar** points that lie on the same plane

**corollary** a proposition that follows directly from a postulate or theorem and can be easily proven

**correlation coefficient** written as  $r$ , describes the strength and direction of the association between two variables; Values for  $r$  range from 1 (perfect positive correlation) to 0 (no correlation) to  $-1$  (perfect negative correlation).

**corresponding angles of figures** the angles that lie in the same position or match up when a transversal intersects two lines

**corresponding sides of figures** the sides that lie in the same position or match up when a transversal intersects two lines

**cosine function**  $\cos x = a$  is a trigonometric function of  $x$  when  $x$  is a radian measure of the angle that intercepts the unit circle at  $(a, b)$

**cosine of an angle** the ratio of the length of the leg that is adjacent to that angle to the length of the hypotenuse; abbreviated  $\cos$

**coterminal angles** angles in standard position that have the same terminal side but have different amounts of rotation; Their angles differ by a multiple of  $360^\circ$ , or  $2\pi$  radians.

**counterexample** an example that shows that a statement is false

**critical z-scores**  $z$ -scores for the upper limit and lower limit of the confidence interval

**cylinder** a solid with two parallel, congruent, circular bases joined by a curved surface

**decreasing function** a function is decreasing within an interval if the value of  $f(x)$  decreases as the value of  $x$  increases

**deductive reasoning** a type of reasoning that uses previously proven or accepted properties to reach conclusions

**degenerate conic section** a conic section formed when a plane intersects the vertex of the double-napped cone

**degree of a monomial** the sum of the exponents of the variable factors

**degree of a polynomial** the degree of the monomial with the greatest degree

**density** the ratio of the mass of an object to its volume;  $\text{density} = \text{mass} \div \text{volume}$

**dependent events** events in which knowing one outcome has an effect on the probability of the other event(s)

**dependent variable** the output variable of a function

**descriptive statistics** statistics that are used to numerically summarize or represent a set of data

**deviation** in a data set, the difference between a data value  $x_i$  and the mean  $\bar{x}$  of the data set;  $x_i - \bar{x}$

**diagonals** line segments with endpoints on nonconsecutive vertices

**diameter** a line segment that connects two points on a circle and contains the center of the circle; also the length of any diameter of a given circle

**dilation** a transformation that changes the size but not the shape of a figure, either by enlarging it or shrinking it

**dimensional analysis** the process of multiplying by conversion factors and dividing out common units

**directed line segment** a segment between two points  $A$  and  $B$  with a specified direction from  $A$  to  $B$  or from  $B$  to  $A$  and a standard distance between the two points

**directrix** one of two references in the definition of a parabola; The distance from any point on the parabola to the directrix (a line) is equal to the distance from that same point to the parabola's focus (a point).

**discrete function** a function with a graph that is disconnected

**discriminant** the radicand  $b^2 - 4ac$  in the quadratic formula

**disjunction** a compound statement that uses the word *or*

**dividend** the number being divided by the divisor

**divisor** the number that divides the dividend

**domain** the set of all allowable inputs for a relation or function

**double-napped cone** cone formed by the generating line that intersects and revolves around another line called the axis; The axis is stationary and the two lines cannot be perpendicular.

**element** a member of a set

**ellipse** a conic section formed when a plane intersects only one nappe, not parallel to the generating line

**end behavior** how a function behaves when the domain values increase or decrease without bound

**endpoints of an interval** the minimum and maximum real numbers on a bounded interval

**endpoints of an intercepted arc** the points where the sides of an inscribed angle meet the circle

**equal complex numbers** complex numbers with real parts that are equal and with imaginary parts that are equal;  $a + bi = c + di$  if and only if  $a = c$  and  $b = d$

**equation** a number sentence that indicates that two expressions have the same value

**equiangular polygon** a polygon with all angles congruent

**equilateral triangle** a triangle that has three congruent sides

**equivalent equations** equations with the same solution or solutions

**equivalent fractions** fractions that have the same value

**equivalent inequalities** inequalities with the same solutions

**even function** a function in which  $f(-x) = f(x)$ ; The graph of an even function is symmetric about the  $y$ -axis.

**event** any particular subset of a sample space

**expanded form of a sum** a representation of a sum as every term added or subtracted to create the sum

**expansion** a dilation for which the absolute value of the scale factor is greater than 1

**experiment** any process that results in one or more results

**explanatory variable** the variable graphed along the horizontal axis in a scatter plot; the independent variable in a statistical analysis

**exponent** in a power, a number used to indicate the number of factors of the base that should be multiplied; For example, in  $5^2$ , 2 is the exponent.

**exponential equation** an equation with variable expressions as exponents

**exponential function** an equation of the form  $f(x) = b^x + k$  where  $b > 0$  and  $b \neq 1$

**expression** a number, a variable, or a combination of numbers, variables, and operations

**exterior angle of a triangle** an angle formed by one side of the triangle and another side of the triangle when it is extended

**extremes** the exterior variables of a proportion

**factor** one of two or more quantities that are multiplied together

**factor of a number** a number that divides into the given number without a remainder

**factorial of a positive integer  $n$**  the product of all the positive integers less than or equal to  $n$ ; written  $n!$

**family of functions** a group of functions with the same fundamental characteristics

**feasible region** the set of all ordered pairs that satisfy the constraints of an optimization problem and are possible solutions to the problem

**Fermi estimate** an estimate of a very large number using the Fermi process; This process involves being able to estimate quantities and then round the estimates to the nearest number that is a product of a counting number less than 10 and a power of 10.

**finite set** a set that has a number of elements that can be described with a whole number

**first quartile,  $Q_1$**  the median of the lower subset of a data set

**five-number summary** the minimum, the maximum, and the three quartiles of a data set

**floor function** another name for the greatest integer function

**flowchart proof** a graphical representation of the logical flow of a proof; In a flowchart proof, statements and conclusions are connected with arrows.

**focus for a parabola** one of two references in the definition of a parabola; The distance from any point on the parabola to the focus (a point) is equal to the distance from that same point to the parabola's directrix (a line).

**FOIL** a mnemonic used for a method to perform the distributive property when multiplying binomials; first-outer-inner-last

**frequency distribution** a table or graph that describes the number of times a value or interval of values occurs in a data set

**frequency of a sinusoidal function** the number of periods per unit; A function's frequency is the reciprocal of its period.

**frequency table** a table that describes the number of times a value or interval of values occurs in a data set

**function** a relation in which each member of the domain is assigned to exactly one member of the range

**function composition** a mapping in which each element of the range of one function is the domain of another function; If  $f$  and  $g$  are functions of  $x$ , the composition of the  $f$  with  $g$  is denoted by  $fg$  and is defined as  $f(g)$ . The domain of  $f(g(x))$  is the set of the domain values of  $g$  with range values that are in the domain of  $f$ .

**function notation** a function equation written so that the dependent variable is replaced with  $f(x)$ ; for example,  $f(x) = 2x + 4$

**generating line of a double-napped cone** the line that revolves about another line, called the axis, to produce a double-napped cone

**geometric sequence** a sequence in which the ratio between consecutive terms is a constant

**geometric series** a series that results from adding the terms of a geometric sequence

**graph of an inequality** a display of all possible solutions of the inequality

**greatest integer function** a function that assigns the greatest integer less than or equal to each real number in an interval; denoted by  $f(x) = \lfloor x \rfloor$  or  $f(x) = \text{int}(x)$ ; also called the floor function

**half-life** the length of time it takes for one-half of a radioactive substance to decay

**half-plane** a plane that has been divided in half by a boundary line

**height** the length of the altitude

**histogram** a bar graph that displays the frequency of data values that occur within certain intervals; The height of each bar gives the frequency in the respective interval.

**horizontal asymptote** the line  $y = b$  for the graph of the function  $f$  if  $f(x)$  approaches  $b$  as  $x$  approaches  $\infty$  or  $-\infty$

**horizontal line test** a test to determine whether a function  $f$  is invertible; If a horizontal line intersects a function more than once, then the function is not invertible. If every horizontal line intersects a function only once, then the function is invertible.

**hyperbola** a conic section formed when a plane intersects both nappes

**hypotenuse** the side opposite the right angle of a right triangle

**hypothesis** the part of a conditional statement that includes the words following *if*

**identity** an equation that is true for all values of the variable; It has infinitely many solutions and is often represented as  $\{x \mid x \in \mathbb{R}\}$ .

**image** a figure after transformation

**imaginary axis** the vertical axis of a complex plane along which the imaginary part of a complex number is graphed

**imaginary number** any number that can be written in the form  $ai$  where  $a$  is any real number and  $i$  is the imaginary unit

**imaginary unit**  $i$  where  $i^2 = -1$  and  $i = \sqrt{-1}$

**improper fraction** a fraction in which the numerator is greater than or equal to the denominator

**incenter of a triangle** the point where the angle bisectors drawn through each vertex intersect

**included angle of a triangle** an angle formed by two sides of a triangle

**included side of a triangle** a side of a triangle that lies between two specific angles

**inconsistent system of equations** a system with no solutions

**increasing function** a function is increasing within an interval if the value of  $f(x)$  increases as the value of  $x$  increases

**independent events** two events in which knowing one outcome has no effect on the probability of the other event

**independent variable** the input variable of a function

**index in sigma notation** the variable  $i$  in sigma notation  $\sum_{i=0}^3 a_i$ ; the variable that takes on all integer values from the lower limit to the upper limit of the summation

**index of a radical**  $n$  in a radical expression  $\sqrt[n]{b}$ ; The index is always greater than 1.

**inductive reasoning** a type of reasoning that starts with observation and moves from a specific observation to a general conclusion

**inequality** a statement formed by placing one of the inequality symbols  $<$ ,  $>$ ,  $\leq$ ,  $\geq$ , or  $\neq$  between two expressions

**inferential statistics** statistics used to draw conclusions or make predictions by taking the information gained from the sample and generalizing it to the population it came from

**infinite set** a set with a boundless number of elements

**initial side of angle** the ray from which the rotation of an angle starts

**inscribed angle** an angle that has its vertex on a circle and its sides as chords of the circle

**inscribed circle** for a given triangle, a circle in the interior of the triangle that touches each side of the triangle at a single point; The incenter is the center of the inscribed circle.

**integers** all the natural numbers, their opposites, and zero; denoted  $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$

**intercepted arc** the arc opposite an inscribed angle

**interior angles of a triangle** the three angles inside the triangle

**interquartile range (IQR)** the difference between the third and first quartiles of a data set;  $IQR = Q_3 - Q_1$

**interval estimate** a range of values that contains the point estimate and is likely to contain the population parameter

**inverse functions** two functions  $f$  and  $g$  that “undo” each other; If you start with a value  $x$ , apply  $f$ , and then apply  $g$ , the result is the original value  $x$ :  $(f \circ g)(x) = x$  and  $(g \circ f)(x) = x$ . The inverse of a function  $f$  is denoted by  $f^{-1}$  (“ $f$  inverse”). For every pair of inverse functions, if  $f^{-1} = g$ , then  $g^{-1} = f$ .

**inverse of a function** a relationship that interchanges the members of the ordered pairs of the original function; denoted by  $f^{-1}$ ; The domain of the inverse function is the range of the original function. The range of the inverse function is the domain of the original function.

**inverse of a relation** a relationship that switches the  $x$ - and  $y$ -values of the ordered pairs of the original relation; The domain of the inverse relation is the range of the original relation. The range of the inverse relation is the domain of the original relation. The graph of the inverse of a relation is its reflection over the line  $y = x$ .

**inverse tangent** the angle that has a given value as its tangent ratio; abbreviated  $\tan^{-1}$

**invertible function** a function  $f$  whose inverse is also a function

**irrational number** a real number that cannot be written in the form  $\frac{a}{b}$  for any integers  $a$  and  $b$

**isometric transformation** any transformation that results in an image that is congruent to the pre-image; also called an isometry

**isosceles triangle** a triangle that has at least two congruent sides

**iterative rule** a rule that can be used to find the  $n$ th term of a sequence without calculating previous terms of the sequence

**lateral edges of a pyramid** the edges of a pyramid that do not form the base; A lateral edge is formed when two lateral faces meet.

**lateral side of an angle** the ray from which the rotation of an angle starts

**lateral surface** the curved surface of a cylinder

**law of detachment** an argument that has two true premises and a valid conclusion; The premises and conclusion have the following form:

premise — If  $a$ , then  $b$ .

premise —  $a$  is true.

conclusion — Therefore,  $b$  is true.

**law of syllogism** a logical argument that always contains two premises and a conclusion; The premises and conclusion have the following form:

premise — If  $a$ , then  $b$ .

premise — If  $b$ , then  $c$ .

conclusion — Therefore, if  $a$ , then  $c$ .

**leading coefficient of a polynomial in simplified form** the coefficient of the first term

**leaf** the digits on the right side of a stem-and-leaf plot

**least common denominator (LCD)** the least common multiple of two or more denominators

**least integer function** a function that assigns the least integer greater than or equal to each real number in an interval; denoted by  $f(x) = \lceil x \rceil$ ; also called the ceiling function

**least squares regression line** the line that makes the sum of the squares of the vertical distances from each data point to the line as small as possible

**length of a line segment** the distance between the endpoints of a line segment

**like radicals** two or more square root expressions that have the same radicand

**like terms** terms that contain the same variable factors taken to the same powers

**line** a collection of points arranged in a straight path

**line of best fit** a least squares regression equation

**line of symmetry** a line over which you can flip a given figure, leaving the figure unchanged; A line of symmetry divides a figure into two congruent (mirror-image) halves.

**line segment** a part of a line that consists of any two points on the line and all the points in between those two points

**line symmetry** a characteristic of a figure in which there is at least one line such that when the figure is folded over the line, the two halves are mirror images that match up perfectly; also called reflection symmetry

**linear equation** an equation whose graph in a coordinate plane is a line

**linear inequality** an inequality that has terms with degree zero or one and an inequality symbol to relate two variables

**linear pair of angles** two angles that have a common side and the same vertex with their other sides point in opposite directions

**linear programming** the process of maximizing or minimizing a linear function subject to a set of conditions, called constraints, that are linear inequalities

**logarithm** the exponent to which a base would have to be raised to result in a given value; The logarithm of  $a$  with base  $b$ ,  $\log_b a$ , where  $b > 0$ ,  $b \neq 1$ , and  $a > 0$ , is defined as  $\log_b a = x$  if and only if  $b^x = a$ .

**logistic growth function** a function of the form

$$f(x) = \frac{C}{1 + Ae^{-Bx}} \text{ where } A, B, \text{ and } C \text{ are all positive constants}$$

**lower limit** in sigma notation, the starting

value for the index  $n$ ; In  $\sum_{n=1}^n a_n$ , the lower limit is 1.

**major arcs** arcs larger than a semicircle

**margin of error** the greatest likely difference between the point estimate and the parameter

**marginal frequency** the ratio of the total for a particular column or row to the overall total in a two-way table

**mean** the statistical average of a data set

**means of a proportion** the middle variables of a proportion

**median of a data set** the middle value when the values are ordered; If the data set has an even number of values, the median is the mean of the two middle values.

**median of a triangle** a segment from the vertex of a triangle to the midpoint of its opposite side

**midline of a sinusoidal function** the horizontal line that is halfway between the maximum and the minimum; The equation of the midline of a sinusoidal function is  $y = \frac{\text{maximum} + \text{minimum}}{2}$ .

**midpoint** a point that divides a line segment into two congruent parts

**minor arcs** arcs smaller than a semicircle

**mixed number** a number consisting of both a whole number and a fraction or the opposite of such a number

**mode** the value that occurs most frequently

**model parameter** a value that is held constant for a specific model

**model variable** a variable that takes on different values for a particular model

**modified box-and-whisker plot** a box-and-whisker plot in which outliers are shown with dots and the whiskers extend to the least and greatest values in the data set that are not outliers

**monomial** a number, a variable, or the product of a number and one or more variables

**multiplication counting principle** if a task can be broken into two stages and there are  $m$  ways of doing the first stage and  $n$  ways of doing the second stage, then there are  $m \cdot n$  ways of doing one thing and the other

**multiplicity** for a root  $a$  of  $p(x) = 0$ , the number of times the factor  $x - a$  occurs in the factorization of any polynomial  $p(x)$

**nappe** one of two equal pieces of a cone where the cone is divided at the vertex by a plane perpendicular to the axis

**natural logarithm** a logarithm with a base  $e$ ; Natural logarithms, such as  $\log_e x$ , are often written using the notation  $\ln x$ .

**natural logarithmic function with base  $e$**   
 $g(x) = \ln x$  where  $x$  is a positive real number; The domain is  $x > 0$ , and the vertical asymptote is  $x = 0$ .

**natural numbers** the set of numbers  
 $\mathbb{N} = \{1, 2, 3, \dots\}$ ; also called counting numbers or positive integers

**nearest integer function** a function that assigns the nearest integer to each real number in an interval; denoted by  $\text{nint}(x)$ ; also called the round function

**negative association** when comparing two data sets, as the data from one set increase, the data from the second set decrease; In a scatter plot, the data points decrease from left to right.

**noncollinear** points that do not lie on the same line

**nonstrict inequality** an inequality that uses  $\leq$  or  $\geq$

**normal distribution** a bell-shaped distribution, centered on the mean

**$n$ th root of  $b$**   $a$  when  $a^n = b$ ,  $a$  and  $b$  are both real numbers, and  $n$  is a positive integer

**number line** a line with equally spaced intervals that are labeled with numbers

**numerical expression** an expression that consists of numbers, operations, and sometimes grouping symbols

**objective function** a linear function that models a quantity that is to be optimized in a linear programming problem

**oblique cone** a cone with an axis that is not an altitude

**oblique cylinder** a cylinder with an axis that is not an altitude

**obtuse angles** angles that measure greater than  $90^\circ$  and less than  $180^\circ$

**odd function** a function in which  $f(-x) = -f(x)$ ; The graph of an odd function is symmetric about the origin.

**one-to-one function** a function in which every element of the domain corresponds to exactly one element on the range

**open half-plane** a half plane in a strict inequality where the boundary line is dashed

**open sentence** an equation or inequality containing one or more variables

**opposite side of an acute angle in a right triangle** the side across from the given angle

**ordered pair** a pair of numbers on a coordinate plane, in which the first number is the  $x$ -coordinate the second is the  $y$ -coordinate

**origin** the intersection of the  $x$ - and  $y$ -axes

**outcomes** the results of an experiment

**outlier** a value far away from most other values in a data set

**parabola** a symmetric curve that is the graph of a quadratic function

**parabola (conic section)** a conic section formed when a plane intersects only one nappe, parallel to the generating line

**parabola (locus of points)** the set of all points in a plane that are equidistant from a fixed line (the directrix) and a fixed point (the focus)

**paragraph proof** a proof in the form of a paragraph

**parallel lines** coplanar lines that never intersect

**parallelogram** a quadrilateral with two pairs of parallel sides

**parameter** a measurement that describes a population

**parent function** the most basic function in a family of functions

**partition a directed line segment** to divide a directed line segment into segments based on a given ratio

**percentile rank** the percentage of data that falls below a particular value

**perfect square** a rational number with a square root that is also rational

**perimeter** the distance around a figure

**period** the length of each interval in a periodic function

**period of a sinusoidal graph** the interval over which a sinusoidal graph repeats itself

**periodic function** a function that repeats itself in regular intervals

**perpendicular bisector** a line, line segment, or ray that passes through the midpoint of a line segment and forms a right angle with the segment

**perpendicular lines** lines that meet at right angles

**piecewise function** a function defined using different rules for different intervals of the domain

**plane** a flat surface that has infinite length and width but no thickness

**point** references a location in space; It has no length, width, or depth.

**point estimates** statistics, such as  $\hat{p}$  and  $\hat{x}$ , that are used to estimate population parameters

**point of tangency** a point where a circle and one of its tangents intersect

**point-slope form of a linear equation** an equation of the form  $y - y_1 = m(x - x_1)$  where  $m$  is the slope and  $(x_1, y_1)$  is a point on the corresponding line

**polygon** a closed figure in a plane formed by three or more line segments, such that each line segment intersects exactly two other line segments at their endpoints only

**polynomial** a monomial or the sum of monomials

**polynomial in  $x$**  a polynomial of the form  $a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$  where the exponents are all whole numbers, the coefficients are all real numbers, and  $a_n \neq 0$

**population** an entire set of members that you want to know something about

**positive association** when comparing two data sets, if the data from one set increase, the data from the second set also increase; In a scatter plot, data points increase from left to right.

**postulates** mathematical statements that are accepted as true without proof

**power** a number that is, or could be, represented by a base with an exponent

**power function** any function that can be written in the form  $f(x) = ax^2 + b$  when  $n$  is a positive integer,  $a$  is any nonzero real number, and  $b$  is any real number

**practical domain and range of a function** the sets of all realistic inputs and outputs for a particular situation

**pre-image** the original figure before a transformation

**premises** statements that are presumed to be true in the course of a logical argument

**prime polynomial** a polynomial that cannot be factored

**principal of a loan** the amount of money actually borrowed

**principal square root** another name for a positive square root; indicated by the radical sign  $\sqrt{\phantom{x}}$

**probability distribution table** a frequency table where each frequency is replaced by the probability of the outcome

**probability of an event** a measure of the likelihood that an event will occur; Probability is always a number between 0 and 1 (inclusive) that can be written as a fraction, a decimal, or a percent.

**product of the functions  $f$  and  $g$**

$$(fg)(x) = f(x) \cdot g(x)$$

**proof** a clear, logical structure of reasoning that begins from accepted ideas and proceeds through logic to reach a conclusion

**proper fraction** a fraction in which the numerator is less than the denominator

**proportion** an equation that states that two ratios are equal; often written as  $a : b = c : d$  or  $\frac{a}{b} = \frac{c}{d}$

**pyramid** a polyhedron with a polygonal base and lateral faces; The faces are triangles that meet at a common vertex.

**Pythagorean identity** for any angle  $\theta$ ,  
 $\sin^2 \theta + \cos^2 \theta = 1$

**quadrantal angles** angles in standard position that have their terminal sides along the horizontal or vertical axis

**quadrants** the sections of a coordinate plane; The first quadrant is between the positive horizontal axis and the positive vertical axis. The second, third, and fourth quadrants are located counterclockwise from the first quadrant, respectively.

**quadratic formula** a formula for finding the solutions of a quadratic equation in the form

$$ax^2 + bx + c = 0 \text{ where } a \neq 0; x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

**quadratic function** a second-degree polynomial function

**quadrilateral** a four-sided polygon

**quotient** the number of times the divisor goes into the dividend evenly

**quotient of the functions  $f$  and  $g$**   $\frac{f}{g}(x) = \frac{f(x)}{g(x)}$ ,  
 $g(x) \neq 0$

**radian measure of a central angle of a circle** the quotient of the angle's arc length and the circle's radius

**radian measure of an angle** the length of the arc on a unit circle subtended (intercepted) by the angle

**radical equation** an equation that contains at least one radical expression with a variable in the radicand

**radical expression** an expression that contains a radical sign  $\sqrt{\phantom{x}}$

**radical function** a function of the form  
 $f(x) = a\sqrt{x-h} + k$  where  $n$  is an integer greater than 1

**radicand** the expression under a radical sign; For example, in the expression  $\sqrt[n]{b}$ ,  $b$  is the radicand.

**radius** a line segment that connects the center of a circle to a point on the circle; also the length of all radii of a given circle

**range of a data set** the difference between its greatest value (maximum) and its least value (minimum)

**range of a relation** the set of possible outputs

**rational expression** a ratio with a numerator and a denominator that are polynomials and with a denominator that is nonzero

**rational function** any function that can be written as the quotient of two polynomials

**rational number** any number that can be expressed as a ratio  $\frac{a}{b}$ , where  $a$  and  $b$  are integers and  $b \neq 0$ ; denoted  $\mathbb{Q}$

**raw score** an original data value

**ray** part of a line that starts at an endpoint and extends infinitely in one direction

**real axis** the horizontal axis of a complex plane along which the real part of a complex number is graphed

**real numbers** the set of numbers that can be written as decimals; the combined set of the rational and irrational numbers; denoted  $\mathbb{R}$

**reciprocal power function** a function that has the power of  $x$  in the denominator of a rational function;

The functions  $f(x) = \frac{1}{x}$ ,  $g(x) = \frac{1}{x^2}$ , and  $h(x) = \frac{1}{x^3}$  are reciprocal power functions.

**rectangle** a parallelogram with four right angles

**recursive rule** a rule for generating terms of a sequence that depends on one or more previous terms of the sequence

**reference angle** for an acute angle  $x$ , the positive acute angle made by the terminal side of  $x$  and the horizontal axis

**reflection** an isometric transformation that flips a figure across a line or line segment, creating a mirror image of the figure

**reflection symmetry** when a figure that has at least one axis of symmetry; The figure can be folded along the axis of symmetry so that both halves match up.

**regression line** a line drawn through the points of a scatter plot to summarize the straight line pattern that the points fit

**regular hexagon** a six-sided polygon with congruent sides and congruent angles

**regular polygon** a polygon that is both equiangular and equilateral

**regular pyramid** a pyramid whose base is a regular polygon and whose lateral faces are congruent isosceles triangles

**relation** a mapping from one set, called the domain, to another set, called the range

**relative frequency** the ratio of the value of a subtotal to the value of the total

**relative maximum point of a function** a point that has a greater function value than all the points on the function that are close to it

**relative minimum point** a point that has a value that is less than all the points on the function that are close to it

**remainder** the amount left over after evenly dividing a dividend by a divisor

**remote interior angles** the angles that are inside the triangle and are not adjacent to a given exterior angle

**repeating decimal** a decimal that does not end but shows a repeating of digits (not made up of all zeros) that goes on forever after the decimal point

**residual** the difference between an observed value and the predicted value from a model; residual ( $e$ ) = observed ( $y$ ) – predicted ( $\hat{y}$ ); the vertical distance between a point on the scatter plot and the point on the linear model directly above or below the point

**residual plot** a graph that shows the residual for each value of the explanatory variable

**response variable** the variable graphed along the vertical axis in a scatter plot; the independent variable(s) in a statistical analysis

**rhombus** a parallelogram with four congruent sides

**right angle** an angle that measures exactly  $90^\circ$

**right cone** a cone with an axis that is also an altitude

**right cylinder** a cylinder with an axis that is also an altitude

**rigid motion** motion that relocates a figure while preserving its shape and size

**roots of a polynomial** the solutions to a polynomial equation in which one side of the equation is a factored polynomial and the other side of the equation is equal to zero

**rotation** an isometric transformation that turns a figure a certain number of degrees, called the angle of rotation, around a central point, called the center of rotation

**rotation symmetry** when a figure can be rotated around its center less than one full turn so that the rotated figure looks exactly like the original figure

**round function** another name for the nearest integer function

**same-side interior angles** angles in between the two lines that are not the transversal and are on the same side of the transversal

**sample** a subset of a population

**sample space** the set of all possible outcomes of an experiment

**scale factor  $t$  of a dilation** the ratio of the length of any side of on an image to the length of its corresponding side on the pre-image

**scalene triangle** a triangle that has no congruent sides

**scatter plot** a graph that displays a set of bivariate data

**secant** a line that intersects a circle in two points

**second quartile,  $Q_2$**  the median of the entire data set that separates the ordered data set into a lower subset and an upper subset

**sector** a region whose boundaries are two radii and part of the circle

**segment bisector** a line, line segment, or ray that passes through the midpoint of a line segment

**semicircle** an arc with endpoints that are also the endpoints of a diameter

**sequence** a function whose domain is the set of natural numbers

**series** the sum of consecutive terms of a sequence

**set** a collection of objects

**sides** the line segments that form a polygon

**sigma notation** a way to write a sum as sum as

$\sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \dots + a_n$ ;  $\sum_{i=1}^n a_p$ , read as “the sum from 1 to  $n$  of  $a_p$ ,” where  $i$  is the index,  $n$  is the upper limit, and 1 is the lower limit; The right side of the equation is the expanded form of the sum.

**similar figures** figures that are the same shape but are not necessarily the same size; The symbol  $\sim$  means “is similar to.”

**similar solids** solids that have the same shape and all the corresponding dimensions are proportional

**similar triangles** two triangles with congruent corresponding angles and proportional corresponding side lengths

**simple event** a single outcome of an experiment; a single element of a sample space

**simplified form of a polynomial** a polynomial that has no like terms, every term is in simplest form, and its terms are in order of decreasing degree with respect to a variable

**simplified radical form of a square root**

**expression** a square root expression in which the radicand is not a fraction, there are no radicals in the denominator, and no factor is a perfect square other than 1

**sine of an angle** the ratio of the length of the leg opposite the angle to the length of the hypotenuse; abbreviated sin

**sine function**  $\sin x = b$  is a trigonometric function of  $x$  when  $x$  is a radian measure of the angle that intercepts the unit circle at  $(a, b)$

**sinusoidal function** a function that has the equation  $f(x) = A \sin Bx + C$  or  $f(x) = A \cos Bx + C$  where  $A \neq 0$  and  $B \neq 0$

**skewed distribution** one side has lower frequencies than the other side

**slope** a number that describes the steepness of a line; the ratio of the vertical change, or rise, to the horizontal change, or run, between any two points on a line

**slope of a line** the ratio of the vertical change, or rise, to the horizontal change, or run, between any two points on the line

**slope-intercept form of a linear equation**

an equation of the form  $y = mx + b$  where  $m$  is the slope and  $b$  is the  $y$ -intercept of the corresponding line

**solution** a value for the variable that makes the equation or open sentence a true statement

**solve** to find all the solutions for an equation

**sphere** the set of all points in space that are a given distance from a point called the center

**square** a parallelogram with four congruent sides and four right angles

**square root** a factor of a number that, when multiplied by itself, results in the number

**standard deviation** a measure of spread of a data set; the standard deviation  $s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$  where  $x$  is a data value,  $\bar{x}$  is the mean of the data set, and  $n$  is the number of data values in the set

**standard form of a linear equation** an equation of the form  $Ax + By = C$  where  $A$ ,  $B$ , and  $C$  are integers, and  $A$  and  $B$  are both nonzero

**standard form of a polynomial** the form of a polynomial in which every term is simplified and its terms are listed by decreasing degree

**standard normal curve** a probability distribution where the mean is 0, the standard deviation is 1, and the total area under the curve is 1

**standard position of an angle** an angle on coordinate plane that has its vertex is at the origin and its initial side along the positive horizontal axis

**statement** a sentence that is either true or false

**statistic** a measurement that describes a sample

**stem** leftmost digit of data values in a stem-and-leaf plot

**stem-and-leaf plot** a data display that lists the last digits (leaves) of the data values to the right of the earlier digits (stems)

**step function** a function defined using a rule that produces a constant value for each designated interval of the domain

**straight angles** angles that measure exactly  $180^\circ$ ; A straight angle is a line.

**strength of association** how closely the data points on a scatter plot fit a straight line pattern

**strict inequality** inequality that uses  $<$  or  $>$

**substitute** to replace

**sum of the functions  $f$  and  $g$**   
 $(f + g)(x) = f(x) + g(x)$

**supplementary angles** angles with measures that sum to  $180^\circ$

**system of equations** a group of two or more equations in the same variables

**system of linear equations** two or more linear equations with the same variables

**system of linear inequalities** a set of two or more linear inequalities using the same variables

**tangent function**  $\tan x = \frac{\sin x}{\cos x}$ ; a trigonometric function of  $x$  in which  $x$  is the radian measure of the angle that intercepts the unit circle at  $(a, b)$ ; The value of  $x$  may not result in the cosine being zero.

**tangent of an angle** the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle; abbreviated  $\tan$

**tangent to a circle** a line in the plane of the circle that intersects the circle in exactly one point

**terminal side of angle** the ray at which the rotation of an angle stops

**terms** the parts of an expression that are added or subtracted

**terms of a sequence** the values of a function whose domain is the set of natural numbers; the range of a sequence

**theorem** a mathematical statement that has been proven to be true

**theoretical domain and range of a function**  
the sets of all allowable inputs and outputs

**third quartile,  $Q_3$**  the median of the upper subset

**transformation** a one-to-one mapping between two sets of points

**translate (a figure)** the sliding of a figure in a straight path without rotating or reflecting it

**translation** a transformation that slides a figure in a straight path without rotation or reflection

**transversal** a line that intersects two or more lines in a plane

**trapezoid** a quadrilateral with exactly one pair of parallel sides

**triangle** a three-sided polygon

**trigonometric functions of angle  $\theta$**   $\sin \theta = \frac{b}{r}$ ,  
 $\cos \theta = \frac{a}{r}$ ,  $\tan \theta = \frac{b}{a}$  ( $a \neq 0$ ) where  $(a, b)$  is a point  
 other than the origin on the terminal side of an angle  $\theta$   
 in standard position, and  $r$  is the distance  $\sqrt{a^2 + b^2}$   
 from the point to the origin

**trigonometric identity** an equation containing  
 a trigonometric ratio that is true for all values of the  
 variable

**trigonometric ratio** the ratio of two sides of a right  
 triangle

**trinomial** a polynomial with three terms

**two-column proof** a proof shown in two columns;  
 The first column shows the steps and the second column  
 shows the justification for each step.

**two-way table** a table that shows data from one  
 sample group as it relates to two different categories

**unbounded interval** the set of all real numbers on  
 one side of a number, called an endpoint; The endpoint  
 may or may not be included.

**uniform distribution** a type of symmetric  
 distribution of data sets where all intervals have the  
 same frequency

**uniform probability distribution** the probability  
 distribution resulting when all values of a random  
 variable  $X$  are equally likely to occur

**unit circle** circle with a radius of one unit

**upper limit** in sigma notation  $\sum_{i=1}^n a_i$ , the variable  $n$ ,  
 which indicates the maximum value for the index  $i$

**valid argument** an argument in which, if the  
 premises are all true, then the conclusion must also  
 be true

**variable** a symbol that represents a value

**variable expression** a combination of variables,  
 numbers, and operations

**variance** a measure of variability of a set relative to  
 its mean; For a data set with values  $x_1, x_2, \dots, x_n$ , the

variance is  $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$  where each difference

$x_i - \bar{x}$  is called a deviation and  $s$  is called the standard  
 deviation

**vertex form of a quadratic function**

$f(x) = a(x - h)^2 + k$  where  $a \neq 0$

**vertex of a double-napped cone** the point where  
 the axis and the generating line of a double-napped  
 cone intersect

**vertex of a parabola** the highest or lowest point on  
 a parabola that opens down or up

**vertex of a polygon** a point where the sides of a  
 polygon meet

**vertex of a pyramid** the common vertex where the  
 triangular faces of a pyramid meet

**vertex of an angle** the common endpoint of two  
 rays that form an angle

**vertex points** corner points of the feasible region

**vertical angles** nonadjacent angles formed by  
 intersecting lines; Vertical angles are congruent.

**vertical asymptote** the line  $x = a$  of the graph  
 of the function  $f$  if  $f(x)$  approaches  $\infty$  or  $-\infty$  as  $x$   
 approaches  $a$ , either from the left or the right

**vertical line test** a test used to determine whether  
 a graphed relation is a function; If the graph is a  
 function, then there is no vertical line that passes  
 through the graph more than once.

**volume** the measure of the space inside (or the  
 space occupied by) a three-dimensional figure;  
 expressed in cubed units

**volume of rectangular prism** the product of the  
 rectangular prism's length, width, and height

**whole number** a number in the set  $\{0, 1, 2, 3, \dots\}$

**x-coordinate** the first number in an ordered pair

**x-intercept** the  $x$ -coordinate of a point where a  
 graph intersects the  $x$ -axis

**y-coordinate** the second number in an ordered pair

**y-intercept** the  $y$ -coordinate of a point where a  
 graph intersects the  $y$ -axis

**zeros of a polynomial function  $f(x)$**  the roots  
 (solutions) of the equation  $f(x) = 0$

**z-score** the number of standard deviations that a  
 data value is from the mean